Laboratory 9 – Exercises

1. Classify the SVHN dataset (32×32 images, 10 classes, 73 257 training images and 26 032 testing images) using a convolutional neural network based on depthwise-separable convolutions. Divide the SVHN training set as follows: 43 257 training images and 30 000 validation images. Use the official test set for final evaluation.

   Define a class implementing the Depthwise-Separable Convolutional Block. For an input with C channels, the block must consist of the following sequence:

   - Depthwise convolution: `Conv2d(C, C, 3 × 3, padding = 1, groups = C)`
   - Pointwise convolution: `Conv2d(C, 2C, 1 × 1)`
   - BatchNorm2d(2C)
   - ReLU activation

   Design the following network architecture:

   (a) An initial convolutional layer: `Conv2d(3, 32, 3 × 3, stride = 1, padding = 1)`, followed by BatchNorm2d and ReLU.

   (b) Four Depthwise-Separable Convolutional Blocks, doubling the number of channels after each block ($32 \to 64 \to 128 \to 256 \to 512$).

   (c) Global average pooling.

   (d) A final Linear layer.

   Train the model for 10 epochs using the Adam optimizer with a learning rate of 0.001 and a batch size of 256. Use the Cross-Entropy loss function during training.

   Produce the following plots:

   - Training accuracy vs. validation accuracy
   - Training loss vs. validation loss

   Finally, compute and report the classification accuracy on the test set.

2. Use the `torchsummary` package to print a detailed summary of the model defined at the previous exercise. The summary must compute the output shape after every layer based on an input of size (3, 224, 224) and must include, for every layer:

   - the output tensor shape,
   - the number of learnable parameters.

   Verify that the output shapes reported by `torchsummary` match the expected dimensions computed manually from the architecture.

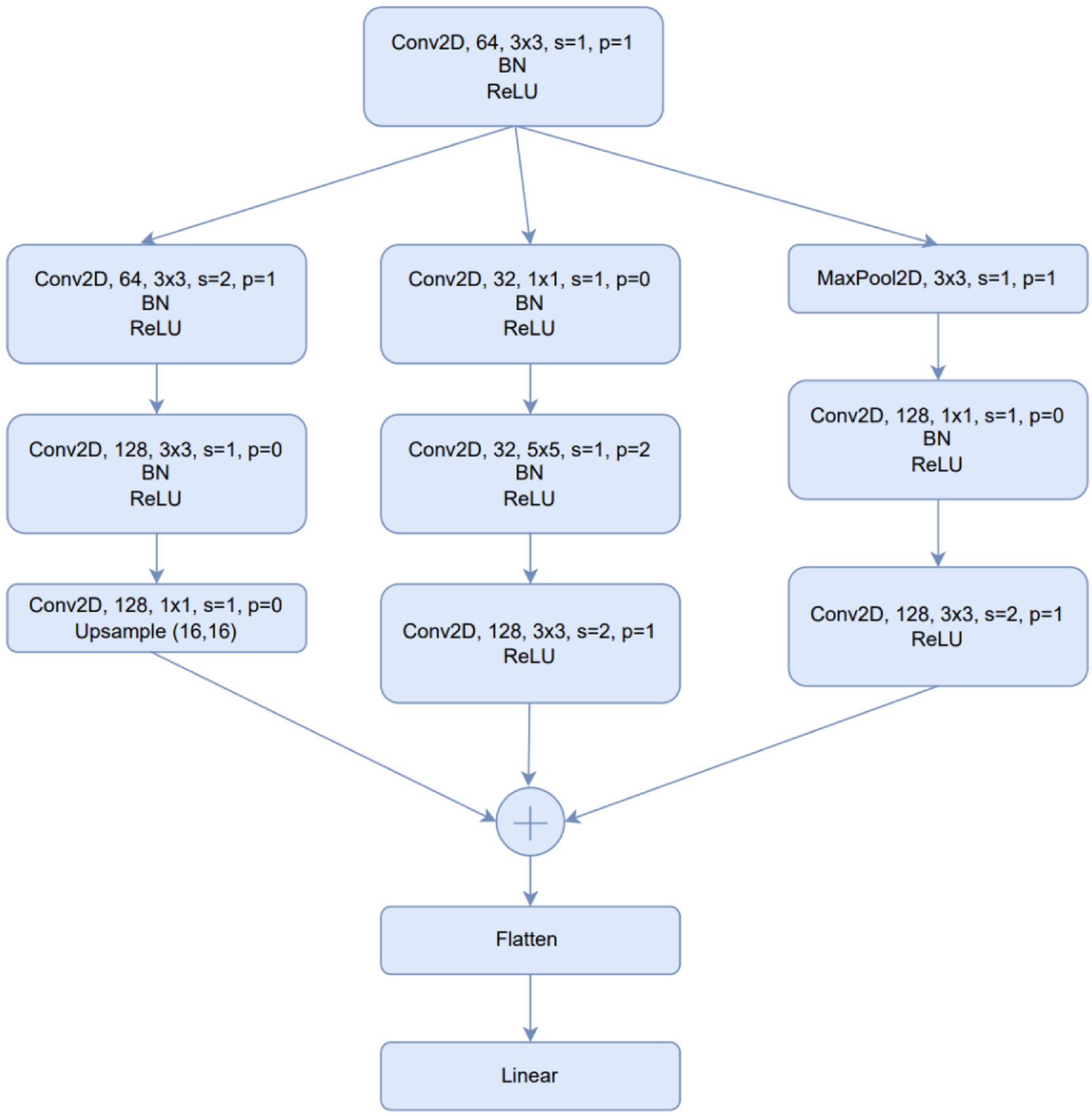

Figure 1: CNN Architecture.

3. Implement the convolutional neural network architecture illustrated in Figure 1. Before training, use the `torchsummary` package to print a complete summary of the model.

   Use the SVHN dataset and split the training set as follows 43 257 training images and 30 000 validation images. Apply data augmentation only to the training subset:

   - Random crop with padding
   - Random rotation
   - Color jitter

   Train the model for 10 epochs using Adam optimizer with learning rate 0.001, batch size 128 and Cross-Entropy loss. Compute the final test accuracy on the SVHN test set and plot the training accuracy vs validation accuracy.